



Efficient, approximate and parallel Hartree–Fock and hybrid DFT calculations. A ‘chain-of-spheres’ algorithm for the Hartree–Fock exchange

Frank Neese^{a,b,*}, Frank Wennmohs^a, Andreas Hansen^a, Ute Becker^b

^a Lehrstuhl für Theoretische Chemie, Institut für Physikalische und Theoretische Chemie, Universität Bonn, Wegelerstr. 12, D-53115 Bonn, Germany

^b Max-Planck Institut für Bioanorganische Chemie, Stiftstr. 12-14, D-45470 Mülheim an der Ruhr, Germany

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ABSTRACT

In this paper, the possibility is explored to speed up Hartree–Fock and hybrid density functional calculations by forming the Coulomb and exchange parts of the Fock matrix by different approximations. For the Coulomb part the previously introduced Split-RI-J variant (F. Neese, *J. Comput. Chem.* 24 (2003) 1740) of the well-known ‘density fitting’ approximation is used. The exchange part is formed by semi-numerical integration techniques that are closely related to Friesner’s pioneering pseudo-spectral approach. Our potentially linear scaling realization of this algorithm is called the ‘chain-of-spheres exchange’ (COSX). A combination of semi-numerical integration and density fitting is also proposed. Both Split-RI-J and COSX scale very well with the highest angular momentum in the basis sets. It is shown that for extended basis sets speed-ups of up to two orders of magnitude compared to traditional implementations can be obtained in this way. Total energies are reproduced with an average error of <0.3 kcal/mol as determined from extended test calculations with various basis sets on a set of 26 molecules with 20–200 atoms and up to 2000 basis functions. Reaction energies agree to within 0.2 kcal/mol (Hartree–Fock) or 0.05 kcal/mol (hybrid DFT) with the canonical values. The COSX algorithm parallelizes with a speedup of 8.6 observed for 10 processes. Minimum energy geometries differ by less than 0.3 pm in the bond distances and 0.5° in the bond angles from their canonical values. These developments enable highly efficient and accurate self-consistent field calculations including nonlocal Hartree–Fock exchange for large molecules. In combination with the RI-MP2 method and large basis sets, second-order many body perturbation energies can be obtained for medium sized molecules with unprecedented efficiency. The algorithms are implemented into the ORCA electronic structure system.

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1. Introduction

In the 1980s Jan Almlöf made several suggestions that revolutionized the application of self-consistent field (SCF) methods to large molecular systems. Almlöf’s first suggestion led to the direct SCF method in which the storage of four-index, two-electron integrals is avoided but instead the integrals are recomputed in every SCF cycle [1]. Negligible contributions to the Fock (or Kohn–Sham, HS) matrix are recognized through pre-screening which, overall, leads to a quadratic scaling algorithm instead of the expected $O(N^4)$. The pre-screening was considerably improved by Häser and Ahlrichs, who pointed out that the Schwartz-inequality holds for two-electron integrals [2]. Ahlrichs and co-workers also generalized this procedure for the direct computation of gradients and Hessian’s [3–5]. This screening has the advantage of great simplic-

ity and efficiency although tighter bounds are possible [6,7], and become advantageous for correlation calculations [7].

While the direct SCF method is now indisputably a cornerstone of modern quantum chemistry, Almlöf’s second suggestion received comparatively less attention. Early on, he recognized that the pre-screening selects fairly different integral batches for the Coulomb and exchange contributions to the Fock matrix [8,9]. Hence, it was reasoned that it might be advantageous to not compute these two contributions together – which seems logical since the same quantities are involved – but to select the best approximation for each contribution [10,11]. This then led to the discovery of linear scaling Fock matrix formation algorithms in which the Coulomb contribution is formed via fast multipole methods [11–20]. The exchange contribution by itself already scales almost linearly and only the pre-screening step scales quadratically with a small pre-factor. In practice scaling laws around $N^{1.5}$ are observed although perfect linear scaling is readily achieved [20–23]. However, as long as the Coulomb and exchange formation algorithms proceed via the generation of traditional two-electron integrals as intermediates several drawbacks still exist: (a) the calculations

* Corresponding author. Address: Lehrstuhl für Theoretische Chemie, Institut für Physikalische und Theoretische Chemie, Universität Bonn, Wegelerstr. 12, D-53115 Bonn, Germany.

E-mail address: neese@thch.uni-bonn.de (F. Neese).

are still fairly expensive, (b) the generation of the two-electron integrals scales unfavorably with the dominant angular momentum in the basis set and (c) the overhead associated with the linear scaling procedures may lead to fairly late crossover with other approximate algorithms.

Following the realization that it might be advantageous to split the Coulomb and exchange treatments Almlöf and Ahmadi developed an efficient, exact, analytic method for the computation of the Coulomb term that was based on the McMurchie–Davidson integral algorithm and consisted of transforming the density to an intermediate Hermite Gaussian basis [10]. Head-Gordon and co-workers later proposed a very similar algorithm which was called the ‘J-engine’ [24]. Both algorithms scaled as $O(N^2)$. In the following, several groups reported linear scaling Coulomb algorithms [11–20].

One of the earliest approximate methods to compute the Coulomb term was invented by Baerends and co-workers in the framework of density functional theory (DFT) [25] with a similar, independent suggestion being made by Whitten [26]. The proposal consists of fitting the electron density to an auxiliary basis set. This reduced the two-electron integral calculation to two- and three-index integrals and reduces the formal scaling of the calculation by an order of magnitude. The physical content of the approximation was carefully analyzed by Dunlap et al. [27–29]. Vahtras, Almlöf and Feyereisen made an important contribution by showing that the most accurate fitting is obtained by minimizing the self-repulsion of the residual [30]. Ahlrichs and co-workers provided an efficient large-scale implementation of the concept (termed ‘resolution of the identity Coulomb approximation’, RI-J) and developed excellent fitting bases for almost the entire periodic table [31,32]. Through these developments it became evident that the error introduced by the fitting is negligible and that the speedups obtained are of a factor of 10–100 [31,32]. Sierka et al. improved the algorithm for large scale application through multipole methods (‘multipole accelerated RI’, MA-RI-J) [33]. Neese combined the Almlöf and Ahmadi concept with the RI-J approximation in the development of the Split-RI-J algorithm that was proven to be very efficient for large and accurate basis sets [34]. The same algorithm was recently re-invented by Head-Gordon and co-workers who also obtained a linear-scaling implementation of the RI concept [35].

An alternative approximation was developed much earlier by Friesner and co-workers (pseudo-spectral approximation, PS) and will be discussed below [36–38].

Thus, as long as the HF exchange is not required (as in ‘pure’ DFT calculations) these methods allow for SCF calculations that are 1–2 orders of magnitude more efficient than traditional HF calculations. It therefore becomes evident that the treatment of the HF exchange is particularly critical for efficient SCF treatments. The nonlocal HF exchange is required in HF and hybrid DFT [39] calculations or as a necessary intermediate in the localized Hartree–Fock [40] method. In addition, if the RI approximation is employed for MP2 (or the improved spin-component-scaled version, SCS-MP2 [41]) calculations, the time needed to solve the SCF equations exceeds by far the time required for the RI-MP2 step despite the formally unfavorable $O(N^5)$ scaling of the latter [42–45]. The dominance of the HF exchange becomes rapidly more pronounced with increasing accuracy of the orbital basis set since the RI-integrals have a more favorable scaling with the L-quantum number than the traditional four-center integrals. Likewise, in RI-MP2 gradient calculations the computationally dominant step involves the parts that proceed via traditional four index integrals (Z-vector equations and integral derivatives required for the SCF part of the gradient) [43,45,46]. This is particularly unfortunate as the recently proposed ‘double-hybrid’ density functionals are gaining popularity [45,47–53]. These functionals have been shown to yield excellent results for energies and many molecular proper-

ties including molecular geometries which became available following the development of analytic gradients [45]. Thus, in order to increase the applicability of hybrid and double-hybrid or RI-MP2 calculations to larger molecules, it is required to improve on the efficiency of the HF exchange treatment.

If the exchange term is computed by the integral direct SCF method, already Almlöf [21,54] has shown that there is only a linear scaling number of contributions to the Fock matrix, provided that Kohn’s conjecture holds that the density matrix elements decay exponentially with distance. While this seems to imply that the HF-exchange is inherently linearly scaling, there still is a quadratically scaling pre-screening step that becomes computationally recognizable in large calculations. In practice, one typically observes a scaling around $N^{1.5}$. Perfectly linearly scaling exchange algorithms that proceed via traditional two-electron repulsion integrals have consequently been developed [20,21,23]. The only drawback of these procedures is that the calculations are still costly. Note also that the algorithm by Ochsenfeld [23] fully exploits the permutational symmetry of the electron–electron repulsion integrals while that of Challacombe [20,21] does not and consequently leads to late crossovers with the standard direct SCF algorithm.

The ORCA package has for almost a decade featured a variant in which the Coulomb term is treated by the RI-J algorithm and the exchange is computed via a standard direct SCF algorithm (it was called RIJONX [55] and frequently used in application studies, e.g. [56–60]; However, as pointed out by a referee RIJONX is an unfortunate name since the exchange treatment is not based on a genuine $O(N)$ procedure. Hence, we refer to this algorithm below as “DX = Direct Exchange”). Even for the largest systems that have been treated so far with several hundred atoms, the time to compute the exchange term strongly dominates the computation time despite its favorable scaling.

Inspired by the success of the RI approximation, several attempts have been made to approximate the HF exchange by similar fitting techniques. Initially, the results of Früchtel et al. in the framework of the NWChem program indicated that efficiency can indeed be gained in this way [61]. This could not be confirmed in later attempts by Neese [62]. Weigend developed an RI-X algorithm (termed RI-K) that proceeded by large scale matrix multiplications and scaled as $O(N^4)$ [63]. It has been shown to be efficient for compact molecules treated with large and accurate basis sets [63]. Polly et al. used the concept of ‘fitting-domains’ to arrive at a linear scaling RI-X algorithm [64]. More recently, further efforts have been reported by Head-Gordon and co-workers [65] as well as by Lindh et al. [66] who used the closely related Cholesky decomposition technique. In both cases speedups relative to the conventional implementation were obtained but the algorithms were not linearly scaling.

A quite different approximation to the two-electron integrals that occur in quantum chemistry has been extensively developed by Friesner and co-workers since the mid 1980s and has been termed the ‘pseudo-spectral’ method [36–38,67–69]. In essence, the pseudo-spectral methods consist of a semi-numeric approximation to the two-electron integrals. From the resulting factorization considerable speedups can be realized for both, the Coulomb and the exchange term. The actual computational machinery, as realized in the Jaguar code, is highly sophisticated and acknowledged to be efficient [67–73]. In order to realize the full potential of the PS-method, the computational algorithms have been optimized for specific basis sets [69–73]. Semi-numerical integration techniques have also been pursued by Termath and Handy [74] and by Van Wüllen [75], both of which tried to approximate the Coulomb term.

The goal of the present paper is the development of an approximate scheme for the computation of the exchange matrix that,

upon combination with the Split-RI-J algorithm for the computation of the Coulomb matrix, leads to an efficient Fock-matrix formation algorithm. The algorithm is related to Friesner's pseudo-spectral algorithm, but differs in a number of aspects from it. In particular, we reasoned that the only drawbacks of the PS methodology are the lack of generality and the high-complexity of the many corrections that are apparently required to approximate the canonical result with sufficient accuracy. Hence, we have focused on a straightforward (and parallel) algorithm that is generally applicable and sufficiently accurate for large-scale applications. We also propose a combination of the RI-approximation with the semi-numeric strategy followed in the pseudo-spectral approaches. Although, the exchange formation will be seen to be potentially linear scaling, the overall computational effort for the formation for the computation of the Fock matrix still scales quadratically due to the quadratic scaling of the Split-RI-J algorithm. However, up to several thousand basis functions, the Coulomb-matrix formation is still considerably more efficient than the computation of the exchange matrix. The algorithm proposed here has a particularly advantageous scaling with the maximum angular momentum in the basis set and is therefore strongest when applied to molecules with up to several hundred atoms in conjunction with large and accurate basis sets.

2. Theory

2.1. Semi-numeric exchange matrix

The aim of this work is to efficiently approximate the elements of an exchange-type matrix:

$$K_{\mu\nu} = \sum_{\kappa\tau} \mathbf{P}_{\kappa\tau} (\mu\kappa|\nu\tau), \quad (1)$$

where $\mathbf{P}$ is some kind of density-type matrix (not necessarily symmetric) and the two-electron integrals are defined over the basis set $\{\varphi\}$ by:

$$(\mu\kappa|\nu\tau) = \int \mu(\mathbf{r}_1) \kappa(\mathbf{r}_1) \nu(\mathbf{r}_2) \tau(\mathbf{r}_2) r_{12}^{-1} d\mathbf{r}_1 d\mathbf{r}_2. \quad (2)$$

The approximation pursued here can be written as follows:

$$K_{\mu\nu} \approx \sum_g X_{\mu g} \sum_{\tau} A_{\nu\tau}(\mathbf{r}_g) \sum_{\kappa} P_{\kappa\tau} X_{\kappa g}. \quad (3)$$

Here the index 'g' refers to grid points $\mathbf{r}_g$ and:

$$X_{\kappa g} = w_g^{1/2} \kappa(\mathbf{r}_g), \quad (4)$$

$$A_{\nu\tau}(\mathbf{r}_g) = \int \frac{\nu(\mathbf{r}) \tau(\mathbf{r})}{|\mathbf{r} - \mathbf{r}_g|} d\mathbf{r}, \quad (5)$$

where w_g denotes the grid weights. Thus, the first integration is carried out numerically and the second one analytically. Note that this destroys the Hermitian character of the two-electron integrals.

Eq. (3) is perhaps best evaluated in three steps:

$$F_{\tau g} = (\mathbf{P}\mathbf{X})_{\tau g}, \quad (6)$$

$$G_{\nu g} = \sum_{\tau} A_{\nu\tau}(\mathbf{r}_g) F_{\tau g}, \quad (7)$$

$$K_{\mu\nu} = (\mathbf{X}\mathbf{G}^+)_{\mu\nu}. \quad (8)$$

As such the equations are very similar to the pseudo-spectral method extensively developed and discussed by Friesner and co-workers since the mid 1980s and commercially available in the Jaguar quantum chemistry package [36–38,70–73,76–78]. The main difference at this point is that instead of $X_{\kappa g}$ there appears a least-square fitting operator $Q_{\kappa g}$ in Friesner's formulation. Further similarities and differences will be discussed below. As

Friesner has pointed out, the evaluation of Eq. (3) scales as $O(N^3)$ [36–38]. Our potentially linear scaling version of Eqs. (6)–(8) is presented below.

2.2. Semi-numeric exchange gradient

Given the exchange matrix, the exchange energy is given by (a sum over spin cases is left out here for simplicity):

$$E_X = \frac{1}{2} \sum_{\mu\nu} P_{\mu\nu} K_{\mu\nu}(\mathbf{P}). \quad (9)$$

Assuming that E_X refers to the nonrelativistic, variational SCF energy, the derivative with respect to parameter λ can be re-arranged to the following form:

$$\frac{\partial E_X}{\partial \lambda} \approx 2 \sum_g \sum_{\mu\nu} \frac{\partial F_{\mu g}}{\partial \lambda} G_{\nu g} \quad (10)$$

with:

$$\frac{\partial F_{\mu g}}{\partial \lambda} = w_g^{1/2} \sum_{\kappa} P_{\kappa\mu} \frac{\partial X_{\kappa g}}{\partial \lambda}. \quad (11)$$

In this formulation, the gradient arises as a minor modification of the exchange matrix formation code. In particular, the derivatives of the analytic integrals are not needed, merely the derivatives of the basis functions on the grid. This has, of course, been clearly realized by Friesner and co-workers in their original gradient implementation that only required the gradient of the least-squares fitting operator but not of the analytic integrals [67].

2.3. Combination of semi-numeric integration with density fitting

As exercised by Friesner and co-workers, the Coulomb matrix can be formed as well through semi-numeric integration by [36–38,68,69,73]:

$$J_{\mu\nu} \approx \sum_g \sum_{\mu'} X_{\mu g} X_{\nu' g} J_g \quad (12)$$

with:

$$J_g = \sum_{\kappa\tau} P_{\kappa\tau} A_{\kappa\tau}(\mathbf{r}_g). \quad (13)$$

Our experience indicates that this approximation is less efficient than the RI-approximation, where [25,29–32]:

$$J_{\mu\nu} \approx \sum_K d_K (\mu\nu|K) \quad (14)$$

Here:

$$(\mu\kappa|K) = \int \mu(\mathbf{r}_1) \kappa(\mathbf{r}_1) K(\mathbf{r}_2) r_{12}^{-1} d\mathbf{r}_1 d\mathbf{r}_2 \quad (15)$$

and $K(\mathbf{r})$ is a member of the auxiliary basis set $\{\eta\}$ that is typically only 2–3 times larger than the orbital basis set [31,32]. Hence, there are many more grid-points than auxiliary basis functions and the evaluation of the Coulomb matrix with semi-numerical integration techniques is more expensive than the RI-approximation. The vector $\mathbf{d}$ represents the density in the auxiliary basis which is best obtained from the linear equation system $\mathbf{V}\mathbf{d} = \mathbf{g}$ where $V_{KL} = (K|L)$ and $g_K = \sum_{\mu\nu} P_{\mu\nu} (\mu\nu|K)$ [30]. The Split-RI-J algorithm [34] was based on the realization that the formation of $J_{\mu\nu}$ and g_K involves many redundant steps and can be improved by transforming $\mathbf{P}$ and $\mathbf{d}$ into an intermediate Hermite Gaussian basis with considerable advantages for large angular momentum containing basis sets [34]. Alternatively, it would be straightforward to improve on the RI algorithm by combining the semi-numeric and density fitting strategies through:

$$g_K \approx \sum_g A_K(\mathbf{r}_g) P_g, \quad (16)$$

$$J_{\mu\nu} \approx \sum_g \sum_{\mu\nu} X_{\mu g} X_{\nu g} J_g, \quad (17)$$

where the intermediates are:

$$P_g = \sum_v X_{vg} F_{vg}, \quad (18)$$

$$J_g = \sum_K A_K(\mathbf{r}_g) d_K. \quad (19)$$

Although we anticipate savings from this treatment compared to the original pseudo-spectral method for the Coulomb matrix, the Split-RI-J algorithm is already so efficient that the Coulomb term was never dominating the computation time in any of our application studies.

3. Implementation

3.1. The chain of spheres algorithm

In order to arrive at an efficient implementation of the semi-numerical exchange treatment, we have developed an algorithm that we have termed the ‘chain of spheres exchange’ (COSX) treatment. The basic idea is simple and consists of constructing small ‘chains’ of shells of basis functions that give non-negligible contributions to the exchange matrix (within the cut-off *Thresh* of the direct SCF procedure). Negligible contributions arise either from negligible differential overlap between pairs of basis functions or from negligible density matrix elements. We refer to the connection of two basis functions by differential overlap as ‘S-junction’ and by a density matrix element by ‘P-junction’. Finding non-negligible S-junctions is achieved by surrounding each contracted basis function by a sphere outside of which the absolute value of the basis function is considered to be negligible (in practice typically $<BFCut = 10^{-10}$). It is emphasized that we do not determine the radius of these spheres from the most diffuse primitive in the contraction but from studying the decay behavior of the entire contracted basis function. Basically, we follow the tail of each basis function in steps of 0.01 Bohr outwards and determine the distance at which $|\phi| < BFCut$. Typically, for a valence basis function of first row-atoms, the spheres have a radius of ~ 3 Å and hence extend over roughly two bonds. Diffuse basis functions obviously have larger sphere radii. Given the spheres, one can construct for each basis function a list of partners with a significant S-junction. Asymptotically, the list is of constant length for each given basis function. Likewise, one constructs for each basis function a list of partners with significant P-junctions that is expected to become asymptotically constant as long as Kohn’s conjecture holds (*vide infra*). Furthermore, taking advantage of the fact that the exchange matrix is linear in the density, the second and later SCF iterations are done with the difference density rather than the full density which greatly diminishes the number of significant P-junctions.

The algorithm proceeds via a loop over batches of grid points. For each batch of grid points there exists a primary set of shells of basis functions (a shell consists of the $2l + 1$ members that share the same radial function) that have non-negligible values of the basis functions at the grid points. These basis functions are denoted $\mu_{\{g\}}$ or $\kappa_{\{g\}}$. Associated with the primary set is a secondary set $\tau_{\{g\}} \equiv \tau_{\kappa\{g\}}$ that is connected via a non-negligible P-junction with the primary shell set. The calculation of the intermediate $F_{\tau_{\{g\}}g}$ then proceeds via BLAS level 3 matrix multiplication of the rectangular matrices $X_{\kappa_{\{g\}}g}$ and $P_{\kappa_{\{g\}}\tau_{\{g\}}}$. Next, the analytical integrals $A_{\nu_{\{g\}}\tau_{\{g\}}}(\mathbf{r})_g$ are calculated and are immediately contracted with $F_{\tau_{\{g\}}g}$ to give $G_{\nu_{\{g\}}g}$. Since the basis functions ν and κ are linked by a S-junction, this is realized via a tertiary list of shells that is con-

nected to $\tau_{\{g\}}$ by a non-negligible S-junction. Since the calculation of the analytic integrals is the rate limiting step of the algorithm, it was coded in an efficient form using hand-optimized integrals for the various angular momentum cases (Scheme 1). This avoids any overhead due to short loops and minimizes expensive storage operations. Finally, the contribution to the exchange matrix is calculated via another BLAS-level 3 operation in which the matrices $X_{\mu_{\{g\}}g}$ and $G_{\nu_{\{g\}}g}$ are contracted. This last contraction amounts to a sum over grid points. Since all matrices used in the intermediate steps are asymptotically of constant size, the entire algorithm is expected to scale linearly with system size. This expectation will be numerically investigated below.

The integral estimate *EST* is computed as a spherical average:

$$EST(\alpha, \beta, d_\alpha, d_\beta) = d_\alpha d_\beta \frac{2\pi}{\alpha + \beta} \exp\left(-\frac{\alpha\beta}{\alpha + \beta} R_{\alpha\beta}^2\right). \quad (20)$$

Here α and β are Gaussian orbital exponents and the d ’s are the contraction factors. The target integral between two s-type functions is obtained from *EST* after multiplication with the incomplete Gamma function F_0 which is confined to values between 0 and 1. Hence *EST* is a conservative upper bound for the integral. The contributions collected in G_{vg} are multiplied by products of $X_{\mu g}$ for which we know that errors are by construction $<BFCUT^2 = 10^{-20}$. Since there are on the order of 10^5 grid points these small individual errors will not accumulate to a large error and perhaps even more aggressive cut-offs would work well in practice. Thus, the entire thresholding procedure is conservative and the accuracy of the approximation is only determined by the size and quality of the integration grid. The efficiency of the algorithm is directly proportional to the number of grid points and the efficiency of the code for the analytic integrals. These subjects will be discussed below.

3.2. Scaling with respect to angular momentum

It is interesting to analyze the computational effort for this method with respect to basis sets that only consist of basis functions with angular momentum L . Friesner and co-workers had already reported that their speedup over conventional algorithms is increasing for more extended basis sets [69]. Here we provide a formal analysis of this finding. We focus on the McMurchie–Davidson (MD) [79] integral algorithm, realizing that the arguments would be somewhat different for alternative, and perhaps more efficient [80], integral treatments. In the MD method, a two-electron integral is calculated as:

$$(\mu\nu|\kappa\tau) = f \sum_{tuw} E_{tuw}^{\mu\nu} \sum_{t'u'v'} (-1)^{t'+u'+v'} E_{t'u'v'}^{\kappa\tau} R_{t+t', u+u', v+v'}. \quad (21)$$

Here $f = 2\pi^{5/2}(\rho\rho'\sqrt{\rho+\rho'})^{-1}$ is a constant that only depends on the four orbital exponents $\alpha, \beta, \gamma, \delta$ as $\rho = \alpha + \beta, \rho' = \gamma + \delta$. The E-coefficients provide the transformation between the Cartesian Gaussian and Hermite Gaussian basis and the R-integrals represent the repulsion integrals between Hermite Gaussian functions. They can all be

```

Loop over secondary basis functions  $\tau_{\{g\}}$ 
  Loop over tertiary basis functions  $\nu$  linked to  $\tau$ 
  Precompute Hermite expansion coefficients E
  Loop over grid points in the current batch
  Calculate integral estimate EST
  Skip if  $|EST * F(\tau, g)| < Thresh$ 
  Calculate  $G_{\nu} A(\tau_g) * F_g$  using hand-optimized routines for
    each angular momentum case
  End loop over batches of grid points
End loop
End loop  $\tau$ 

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Scheme 1. Loop structure for the computation of the key intermediate $G_{\nu_{\{g\}}g}$. In the actual code the permutation symmetry of ν and τ is also considered.

generated by recursive algorithms [79,80]. Importantly, the loops over tu and $t'u'$ are all of a length proportional to L such that the overall scaling of the integrals is $O(L^{10})$ and this carries over to the exchange matrix as well. By contrast, in the pseudo-spectral and COSX algorithms, the steepest scaling intermediate (with respect to L) is G_{vg} that shows an $O(L^5)$ behaviour. Thus, it is evident that once the higher angular momentum basis functions start to dominate the computational effort the advantages of the semi-numeric algorithms over the conventional treatments become very large. In practice this already occurs for the popular triple- ζ basis sets with 2d1f polarization sets on the heavy atoms and 2p1d on the hydrogens (e.g. TZVPP, or 6-311G(2df,3pd)).

3.3. Comparison to the pseudo-spectral method

In order to arrive at the smallest possible grids, Friesner et al. have implemented a number of sophisticated numerical methods [36–38,67–69,73]. Foremost, in order to realize a truly pseudo-spectral method (as opposed to the semi-numerical treatment pursued here) it is necessary to introduce a least squares fitting operator Q_{kg} that replaces X_{kg} in the second basis function of the bra. Note that this breaks the permutation symmetry of the basis function product in the bra. In order to evaluate this operator it is furthermore necessary to introduce a set of ‘dealiasing’ functions that supplement the orbital basis set. The set of dealiasing functions is, however, specific for each orbital basis set and must be carefully optimized in order to obtain accurate results [36–38,67–69,73]. It is then necessary to carefully analyze the orbital basis set in terms of short-range and long-range components in order to not destroy the locality that is inherent in X_{kg} [69]. Secondly, Friesner and co-workers have chosen to correct for the errors of the PS method by computing parts of the integrals (one-center, two-center and even part of the three-center integrals) by analytical means [69]. As they rightly point out, this has to be done in a careful way in order to not add significantly to the computation time. It does, however, take some of the favorable L -scaling inherent in the semi-numerical method away. The required analytic integrals are different for the Coulomb and the exchange part of the Fock matrix [69]. Finally, Friesner et al. have carefully optimized different grids and have developed specific iterative schemes in which different grids are used in different SCF cycles in order to further increase the efficiency of their algorithm [69].

It is indeed impressive what has been achieved along these lines. However, in this work, we do have a slightly different focus. We do not want to deal with the intricacies of dealiasing functions; basis set dependent integration grids or geometry dependent integral corrections. All of this complicates the implementation of gradients and presumably leads to high code complexity including more complex parallelization. After considerable testing, we have decided to only keep one-center corrections that can be introduced at negligible computational cost and that increase the stability of the results with respect to the size of the integration grid. Even those corrections are not without pitfalls. If one integrates the one-center integrals on purely atomic grids one can calculate the corrections very efficiently, since they become identical for all atoms that share the same basis set. However, one introduces an additional error by such a procedure since the atomic grids are ‘distorted’ by the molecular environment through the Becke construction scheme [81]. The computation of the one-center corrections on the molecular grid, however, becomes more challenging and, according to our experience, does not lead to considerably better results. We thus prefer the simple one-center corrections on atomic grids.

In essence, we accept to pay the price of somewhat larger integration grids and demonstrate below that large computational savings can still be realized for arbitrary basis sets without

significantly sacrificing the accuracy of the calculation. In more recent works, for example in the framework of TD-DFT [70], Friesner and co-workers have apparently also followed such a strategy. Secondly, we prefer to treat the Coulomb term via the Split-RI-J approximation rather than the semi-numerical treatment. The resulting combination of approximations is termed RIJCOSX and has been coded into a development version of the ORCA program package. Since both, the COSX and Split-RI-J algorithms behave very well with respect to basis set size, efficient and accurate calculations with large basis for large molecules become feasible.

4. Numerical results

4.1. Computational details

All calculations have been performed with a development version of the ORCA program [82] in combination with the split-valence (SV [83]), triple- ζ valence (TZV [84]) and quadruple- ζ (QZV [85]) basis sets developed by the Karlsruhe group that were supplemented with the appropriate polarization functions from the TurboMole library [86]. For the fitting basis sets in the RI-J treatment, the ‘def’ fit bases optimized for the SV, TZV and QZV basis sets were used but truncated after $L = 4$ due to technical constraints. Calculations labeled RIJDX used exact integration for the exchange contribution and the Split-RI-J approximation for the Coulomb term. Calculations labeled RIJCOSX used the COSX exchange approximation in conjunction with Split-RI-J. Calculations were run on 2.6 GHz Dual Core Athlon machines. All reported times are wall-clock times and have a variability of about 10–15% due to differing system load. For DFT calculations, the standard integration grids of the ORCA package were used. All calculations were converged to 10^{-7} Eh in the total energy (ORCA keyword TightSCF).

4.2. Integration grids

In the first part of the development, we have extensively tested various integration grids that were derived from the standard DFT grids included in the ORCA package. The reference point is a restricted HF (RHF) calculation with the RIJDX approximation. The accuracy of the integration grids is controlled by the radial resolution parameter ‘ ϵ ’: the number of radial shells n_r for a given atom is calculated as [87]: $n_r = 15\epsilon + 5r - 40$, where r is the row of the periodic table to which the atom belongs) together with the largest Lebedev grid used in constructing the atomic grid. The pruning algorithm of Gill et al. is used to reduce the number of points in the inner and outer regions [88].

We choose to present the results obtained for four molecules with diverse basis sets in Fig. 1. Cyclohexane represents a saturated molecule, Guanine a planar π -system, S_8 introduces third row atoms and CF_3-COO^- represents an anionic system requiring diffuse basis functions. It is obvious that arbitrary accuracy can be obtained in the limit of infinitely large integration grids. In order to arrive at a reasonable compromise between computational cost and accuracy it is therefore necessary to define a reasonable accuracy goal. We define this goal as an error of about 0.1 kcal/mol in the total exchange energy. This would ensure that energy differences are reliable to better than 1 kcal/mol, the desired chemical accuracy. It is obvious from Fig. 1 that this type of accuracy is achieved with a 194 point angular grid and a radial integration accuracy of $\epsilon = \frac{13}{3}$. With the smaller 50 point and 110 point grids one reliably reaches an accuracy of ~ 1 kcal/mol with $\epsilon = 4$. Interestingly, (except for CF_3-COO^-) the 110 point grid is only a moderate improvement over the 50 point grid and the 302 point grid is only a moderate improvement over the 194 point grid. Hence,

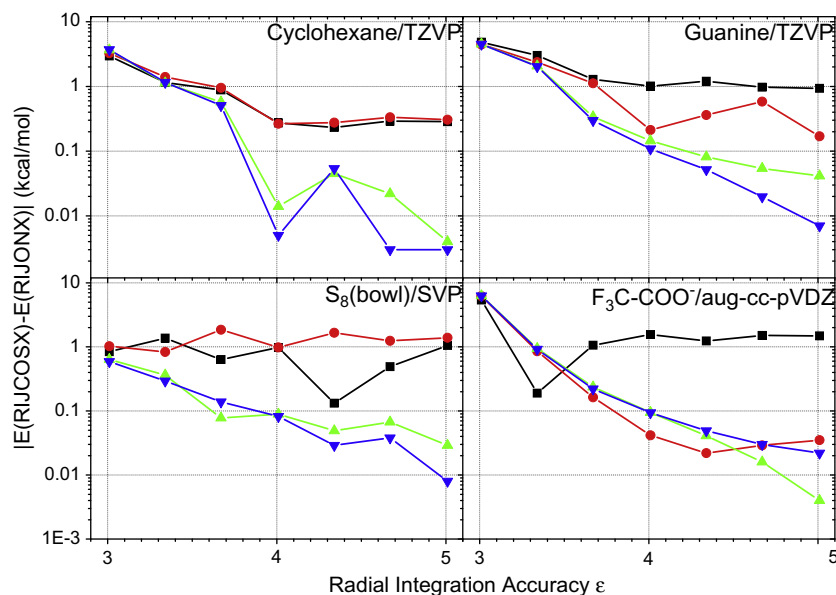


Fig. 1. Convergence of the COSX error with the radial integration accuracy for a variety of molecules/basis sets and angular grids. Squares = 50 point Lebedev grid, circles = 110 point Lebedev grid, up-pointing triangles = 194 point Lebedev grid and down-pointing triangles = 302 point Lebedev grid. The error is defined as the difference in the final converged SCF energies for the RJCOSX approximation relative to RJDX.

the smaller grids have been given preference in defining the standard grid for chemical applications.

After considerably testing, we hence propose the following strategy: (a) carry out the initial SCF iterations (up to a DIIS error of $1e-3$ or the start of the SOSCF [89,90] procedure) with a small 50 point grid and $\varepsilon = \frac{10}{3}$. This leads to about 300–450 points per atom; (b) converge the SCF with a 50 point grid and $\varepsilon = 4$. This leads to 500–600 grid points per atom; (c) recompute the exchange energy on a 194 point Lebedev grid with $\varepsilon = \frac{13}{3}$ (~2000 points per atom). Gradients are computed on the largest grid.

We note that the final exchange energy can be computed considerably more efficiently than the exchange matrix since many more of the costly analytic integrals can be eliminated in the prescreening. In practice, at least a factor of two in the computa-

tion time can be saved if only the energy rather than the whole exchange matrix is computed. This largely compensates for the use of a larger final grid. A similar observation has been made by Almlöf long ago in the context of the direct SCF procedure. However, Almlöf also realized that screening the energy leaves an unmonitored error in the Fock matrix that would prevent convergence [54].

If required, a series of default grids with increasing accuracy has been defined. For example, for hybrid DFT calculations with only 20% exact exchange (e.g. B3LYP) it is typically sufficient to recompute the final energy with $\varepsilon = \frac{14}{3}$ and the 110 point Lebedev grid instead of the more costly 194 point grid. A similar strategy with a final energy evaluation on a larger grid has been followed with success for almost a decade in the ORCA DFT code and has probably

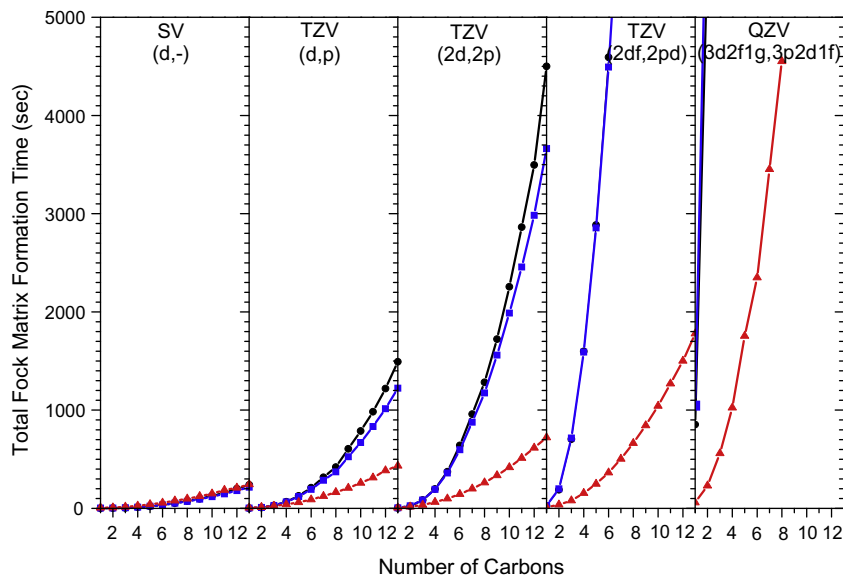


Fig. 2. Comparison of the efficiency of the RJDX and RJCOSX approximations relative to the parent direct RHF method for linear carbon chains $H_3C-(CH_2)_n-CH_3$ as a function of basis set. Circles: approximation free RHF, Quadrats: RJDX, Triangle: RJCOSX. Plotted is the total Fock matrix formation time summed over all SCF iterations. Typically 10–11 iterations were needed for convergence.

Table 1
Accuracy and efficiency of the RIJDX and RIJCOSX algorithms for RHF calculations on medium sized molecules with different basis sets. All calculations were done with the default grids and basis set labels refer to the 'def2' variants [92]. The speedup refers to the ratio of total wall clock times required to finish the calculation.^a

Molecule	N_{Atoms}	Basis	N_{Basis}	Δ_{RIJDX} (kcal/mol)	Δ_{RIJCOSX} (kcal/mol)	Speedup (RIJDX)	Speedup (RIJCOSX)	t_{RIJCOSX} (sec) ^b
Vancomycin	176	SV(P)	1572	−2.9	−1.1	1.7	5.0	15278
		TZVPP	4197					121656
Morphin	40	SV(P)	332	−0.6	−0.3	1.2	3.1	726
		TZVP	765	−0.5	0.3	1.3	30.7	4907
		TZVPP	917	−0.6	0.4	1.2	32.7	5644
Beclomethasone	57	SV(P)	454	−0.7	−0.2	1.4	3.4	1501
		TZVP	1045	−0.7	0.4	1.6	37.9	9365
		TZVPP	1282	−0.7	0.6	1.1	27.3	11229
Captopril	30	SV(P)	220	−0.5	−0.2	1.1	1.5	278
		TZVP	508	−0.4	0.1	1.2	15.6	1696
		TZVPP	649	−0.5	0.1	1.2	18.7	2131
Epinephrine	26	SV(P)	208	−0.4	−0.3	1.1	1.6	217
		TZVP	481	−0.4	0.0	1.1	15.4	1429
		TZVPP	585	−0.4	0.1	1.2	18.6	1684
		QZVP	1131	−0.5	−0.4	1.2	40.5	13482
Flutamide	30	SV(P)	288	−0.5	−0.4	1.1	2.2	385
		TZVP	655	−0.4	0.3	1.2	20.4	2566
		TZVPP	743	−0.4	0.3	1.2	21.2	2876
Nitroglycerine	17	SV(P)	178	−0.4	−0.5	1.1	1.6	133
		TZVP	402	−0.3	−0.2	1.1	15.1	840
		TZVPP	442	−0.3	−0.2	1.3	16.6	932
		QZVP	834	−0.3	0.2	1.2	33.5	7305
Penicillin	42	SV(P)	376	−0.8	−0.3	1.2	2.5	671
		TZVP	855	−0.7	0.0	1.3	22.2	4760
		TZVPP	1004	−0.7	0.1	1.1	19.0	5394
Tetracycline	56	SV(P)	496	−0.9	−0.5	1.4	3.5	1595
		TZVP	1136	−0.8	0.1	1.3	30.5	10466
		TZVPP	1328	−0.9	0.2	1.4	35.0	11619
Cholesterol	74	SV(P)	484	−0.7	−0.3	1.2	2.4	1563
		TZVP	1144	−0.7	0.4	1.2	24.3	9630
		TZVPP	1512	−0.7	0.7	1.6	33.8	12420
Menthol	31	SV(P)	194	−0.3	−0.2	1.1	1.4	230
		TZVP	461	−0.3	−0.1	1.2	15.4	1292
		TZVPP	621	−0.3	0.0	1.2	19.2	1724
		QZVP	1227	−0.4	−0.2	1.4	61.2	14263
Tryptophane	27	SV(P)	234	−0.4	0.0	1.1	1.9	296
		TZVP	537	−0.3	0.3	1.2	17.1	1900
		TZVPP	633	−0.4	0.4	1.2	21.8	2179
		QZVP	1215	−0.4	−0.3	1.2	55.0	17481
Histidine	20	SV(P)	172	−0.3	−0.1	1.1	1.3	135
		TZVP	395	−0.3	0.1	1.1	14.8	858
		TZVPP	467	−0.3	0.1	1.2	17.3	1004
		QZVP	897	−0.3	−0.3	1.2	45.5	8106
Cysteine	14	SV(P)	116	−0.3	−0.2	1.0	1.0	60
		TZVP	395	−0.2	−0.1	1.1	10.8	345
		TZVPP	467	−0.3	−0.1	1.0	11.7	417
		QZVP	897	−0.3	−0.1	1.2	37.3	3690
Glycine	10	SV(P)	80	−0.2	−0.2	0.8	0.6	30
		TZVP	185	−0.1	0.0	1.1	7.8	142
		TZVPP	225	−0.2	0.0	1.1	9.5	172
		QZVP	435	−0.2	−0.1	1.0	25	1583
Tyrosine	24	SV(P)	204	−0.4	−0.1	1.1	1.5	207
		TZVP	469	−0.3	0.2	1.2	16.7	1272
		TZVPP	557	−0.4	0.2	1.2	19.5	1488
		QZVP	1071	−0.3	−0.3	1.1	44.1	12237
Adenine	15	SV(P)	150	−0.3	0.0	0.8	1.4	99
		TZVP	340	−0.2	0.1	1.1	16.4	696
		TZVPP	380	−0.3	0.1	1.1	17.6	754
		QZVP	720	−0.3	−0.2	1.1	45.5	5726
Cytosine	13	SV(P)	122	−0.3	−0.1	1.0	1.3	64
		TZVP	278	−0.2	0.1	1.1	12.8	425
		TZVPP	318	−0.2	0.1	1.1	14.5	474
		QZVP	606	−0.2	−0.2	1.1	38.4	3832
Guanine	16	SV(P)	164	−0.3	−0.2	1.0	1.7	119
		TZVP	371	−0.3	0.0	1.0	14.1	831
		TZVPP	411	−0.3	0.0	1.1	17.5	887
		QZVP	777	−0.3	−0.2	1.2	45.9	7363
Thymine	15	SV(P)	138	−0.3	−0.2	1.0	1.3	86
		TZVP	315	−0.2	0.0	1.1	13.6	583
		TZVPP	363	−0.3	0.1	1.1	15.9	634
		QZVP	693	−0.2	−0.2	1.2	41.0	5051
D-Glucose	24	SV(P)	192	−0.4	−0.5	1.1	1.5	184
		TZVP	444	−0.4	−0.3	1.3	16.3	1128

Table 1 (continued)

Molecule	N_{Atoms}	Basis	N_{Basis}	Δ_{RIJDX} (kcal/mol)	Δ_{RIJCOSX} (kcal/mol)	Speedup (RIJDX)	Speedup (RIJCOSX)	t_{RIJCOSX} (sec) ^b
CO-Heme	86	TZVPP	540	−0.4	−0.2	1.3	18.2	1383
		QZVP	1044	−0.4	−0.4	1.3	47.4	11 283
		SV(P)	789	−1.2	−0.1	1.5	4.0	6814
		TZVP	1780	−1.1	1.6	1.7	38.9	42 996
Ferrocene	21	TZVPP	2087	−1.2	1.8	1.9	41.7	51 005
		SV(P)	191	−0.3	−0.2	1.1	2.9	479
		TZVP	415	−0.2	0.3	1.3	13.8	3157
		TZVPP	514	−0.3	0.4	1.2	14.1	4256
Ni(tren) ₂ (NCS) ₂	35	QZVP	954	−0.3	−0.5	1.2	42.2	26 825
		SV(P)	299	−0.6	−0.1	1.2	2.0	2123
		TZVP	655	−0.5	0.5	0.9	10.7	13 585
		TZVPP	828	−0.6	−1.8	1.3	10.2	25 195
Dibenzo-Crown18.6	50	SV(P)	412	−0.8	−0.5	1.3	2.3	789
		TZVP	950	−0.7	0.0	1.3	19.1	5226
		TZVPP	1142	−0.8	0.0	1.3	21.9	6121
Mean error				−0.46	−0.03			
Means abs. error				0.46	0.27			
Min. error				−2.9	−1.8			
Max. error				−0.1	1.8			

^a For the QZVP calculations a 302 point final grid has been used.

^b t_{RIJCOSX} is the total wall clock time required to finish the entire SCF calculation including the initial guess and all overhead.

first been suggested and extensively used by the TurboMole developers [91].

4.3. Benchmark results: accuracy and efficiency

In order to study the efficiency of the COSX implementation we have initially studied hydrocarbon chains $\text{H}_3\text{C}-(\text{CH}_2)_n-\text{CH}_3$ as a function of chain length and basis set. For the small SV(d) basis set RHF (meaning standard closed shell direct SCF), RHF-RIJDX and RHF-RIJCOSX all perform very similarly. Here the systems are just too small for the approximation to become effective. The crossover between the standard direct SCF procedure and RIJCOSX occurs at $\text{C}_{12}-\text{C}_{13}$. Thus, for large molecules, the lower scaling of the COSX procedure starts to pay off. Much more significant are the savings for larger basis sets. Already for the rather moderate TZV(d,p) basis set the savings for C_{13} are already a factor of two and after adding the second set of polarization functions increase to a factor of four. After adding *f*- and *g*-functions the RIJCOSX savings become dramatic and approach a factor of fifty. It is particularly pleasing that the increase in computation time for RIJCOSX is very moderate upon going from SV(d) to TZV(d,p) to TZV(2d,2p) to TZVPP. The RIJCOSX calculations only become somewhat expensive for the extensive QZV(3d2f1g,3p2d1f) basis set. Even for the largest calculations, the Split-RI-J part of the calculation takes much less time (factor 5–10) than the COSX procedure thus demonstrating the extremely high efficiency of this approximation. The savings offered by RIJDX appear very limited in this set of test calculations but do approach a factor of two in our experience for larger systems (see Fig. 2).

In order to demonstrate the efficiency and accuracy of the RIJCOSX approximation in ‘real-life applications’ we have put together a test set of 26 medium sized molecules in the range 15–174 atoms and up to ~2000–4000 basis functions (Table 1).

The data in Table 1 demonstrate that the RIJDX approximation provides speedups of less than a factor of two and introduces an error that is always negative. This makes sense of course, since the RI-J approximation always underestimates the true Coulomb energy. The mean error is ~0.5 kcal/mol and only for molecules as large as Vancomycin (176 atoms) exceeds 1 kcal/mol.

The RIJCOSX approximation is similarly accurate but shows errors that are either positive or negative. The trend is to errors of positive sign for more extended basis sets. Importantly, the error remains small for extended basis sets up to TZVPP. For the very flexible QZVP basis set we have preferred to revert to a 302 point

final grid and in this case the absolute errors remain very small. There tends to be a cancellation of errors from the RI-J and COSX approximations such that the mean absolute RIJCOSX (0.27 kcal/mol) errors are smaller than the RIJDX ones (0.46 kcal/mol).

The speedups observed for RIJCOSX range from moderate for small basis sets such as def2-SV(P) (factors of 1–5) and become spectacularly large for extended basis sets such as TZVPP and QZVP (up to a factor of 60). Thus, the RIJCOSX approximation is highly attractive for fast and accurate calculations on large molecules.

4.4. Combination with RI-MP2

In order to investigate whether the RIJCOSX wavefunction is accurate enough to be combined with the RI-MP2 method, we have carried out some further test calculations on some of the molecules in Table 1 with the TZVPP basis set (Table 2). On average, a speedup of a factor of ~15 is obtained for the completion of the entire calculation. The errors range from 0.01 to 1.2 mEh and thus are thus still within chemical accuracy of the standard RI-MP2 result. Hence, it is highly beneficial to combine RIJCOSX with RI-MP2, RI-SCS-MP2 or double hybrid DFT.

4.5. Benchmark results: reaction energies

Since total energies are of little concern for chemistry, we have also tested the accuracy of the RIJDX and RIJCOSX approximations on the G2/97 set [93] supplemented by some closed shell reaction energies for larger molecules proposed by Grimme [94]. The results in Table 3 demonstrate that the RIJDX approximation only introduces an error of well below 0.1 kcal/mol which is therefore negligible. The RIJCOSX approximation introduces an error of only

Table 2

Accuracy and efficiency obtained from the combination of RIJCOSX with RI-MP2.^a

Molecule	N_{Atoms}	N_{Basis}	Error (mEh)	Speedup
Tetracycline	56	1328	0.15	14.3
Morphine	40	917	1.21	18.9
Beclomethasone	57	1285	0.65	15.9
Captopril	30	652	0.01	12.1
Peniciline	42	1007	0.46	13.5

^a The speedup is defined as the ratio of the wall clock times for the standard RI-MP2 and the RIJCOSX-RI-MP2 calculation. Since the RI-MP2 step is identical for both sets of calculations the entire speedup results from the savings in the SCF part.

Table 3

Statistical evaluation of reaction energies for 20 closed shell reactions from the G2 set as well as additional 9 reactions for larger molecules. Reported are the errors of the RIJDX and RIJCOSX approximations relative to the parent RHF/TZVP and B3LYP/TZVP methods using the standard grids. All energies are in kcal/mol.

Reaction	E_{reac}	Error		E_{reac}	Error	
	RHF	Δ_{RIJCOSX}	Δ_{RIJDX}	B3LYP	Δ_{RIJCOSX}	Δ_{RIJDX}
1. $\text{H}_2 + \text{F}_2 \rightarrow 2\text{HF}$	-144.07	0.26	0.09	-127.63	0.09	0.07
2. $\text{OF}_2 + \text{H}_2 \rightarrow \text{F}_2 + \text{H}_2\text{O}$	-76.29	-0.01	0.04	-62.71	0.02	0.03
3. $\text{O}_3 + 3\text{H}_2 \rightarrow 3\text{H}_2\text{O}$	-271.53	-0.05	0.09	-216.11	0.04	0.07
4. $\text{N}_2 + 3\text{H}_2 \rightarrow 2\text{NH}_3$	-33.95	0.14	0.12	-38.00	0.09	0.09
5. $\text{N}_2\text{O} + \text{H}_2 \rightarrow \text{N}_2 + \text{H}_2\text{O}$	-103.62	-0.04	0.00	-74.03	-0.01	0.00
6. $\text{C}_2\text{H}_2 + \text{H}_2 \rightarrow \text{C}_2\text{H}_4$	-50.65	0.17	0.04	-49.77	0.06	0.03
7. $\text{CO} + \text{H}_2\text{O} \rightarrow \text{CO}_2 + \text{H}_2$	1.02	0.03	0.00	-13.08	0.01	0.00
8. $\text{CO}_2 + \text{H}_2\text{O} \rightarrow \text{HCOOH}$	-2.23	0.02	0.02	-0.51	0.01	0.02
9. $\text{C}_2\text{H}_2 + \text{HF} \rightarrow \text{H}_2\text{C}-\text{CHF}$	-26.99	-0.10	-0.01	-30.20	-0.03	-0.02
10. $\text{H}_3\text{C}-\text{CHO} \rightarrow \text{CO} + \text{CH}_4$	-11.87	0.07	0.01	-1.03	0.00	0.00
11. $\text{C}_2\text{H}_2 + \text{C}_2\text{H}_4 \rightarrow \text{cyclobutene}$	-23.31	-0.11	-0.01	-27.24	-0.03	0.00
12. $\text{C}_2\text{H}_4 + 4\text{F}_2 \rightarrow \text{C}_2\text{F}_4 + 4\text{HF}$	-466.44	0.39	0.17	-416.68	0.12	0.12
13. $\text{SiCl}_4 + \text{CF}_4 \rightarrow \text{SiF}_4 + \text{CCl}_4$	8.40	0.28	0.07	-0.51	0.06	0.06
14. $\text{BCl}_3 + \text{AlF}_3 \rightarrow \text{BF}_3 + \text{AlCl}_3$	-48.90	-0.41	-0.02	-36.96	-0.04	-0.02
15. $\text{F}_2 + \text{Cl}_2 \rightarrow 2\text{ClF}$	-27.38	-0.07	-0.02	-23.84	-0.02	-0.01
16. $\text{CH}_2\text{O} + \text{H}_2 \rightarrow \text{CH}_3\text{OH}$	-28.01	0.09	0.04	-26.24	0.03	0.03
17. $\text{P}_2 + 3\text{F}_2 \rightarrow 2\text{PF}_3$	-493.26	0.43	0.07	-447.19	0.06	0.07
18. $\text{Li}_2 + \text{F}_2 \rightarrow 2\text{LiF}$	-199.14	-0.08	0.07	-195.65	0.02	0.05
19. $\text{SO}_2 + 3\text{H}_2 \rightarrow \text{H}_2\text{S} + 2\text{H}_2\text{O}$	-111.27	-0.01	0.09	-82.51	0.04	0.06
20. $\text{CO} + 3\text{H}_2 \rightarrow \text{H}_2\text{O} + \text{CH}_4$	-58.55	0.16	0.12	-63.77	0.10	0.10
21. $\text{C}_{20}(\text{bowl}) \rightarrow \text{C}_{20}(\text{cage})$	63.77	0.46	-0.11	41.91	-0.03	-0.08
22. Heptahexane $\rightarrow$ heptatriyne	-22.77	-0.03	-0.01	-6.06	-0.02	-0.02
23. 2,2,3-Dimethylbut-2-ene $\rightarrow$ octamethylcyclobutane	3.79	-0.51	0.10	5.37	-0.07	0.08
24. 2-Hydroxypyridine $\rightarrow$ 2-hydropyridone	1.46	0.02	0.00	-1.18	0.00	0.00
25. $\text{C}_2\text{H}_4 + \text{H}_2\text{CN}_2 \rightarrow 1\text{-pyrazoline}$	-35.49	-0.38	-0.13	-27.47	-0.19	-0.12
26. 2p-Xylene $\rightarrow$ [2.2]paracyclophane + 2H ₂	89.12	0.50	-0.15	76.16	-0.03	-0.13
27. $\text{C}_{12}\text{H}_{12}$ (D _{6h} cage) $\rightarrow$ $\text{C}_{12}\text{H}_{12}$ (cyclopentadienyletropenyle)	10.04	0.17	0.03	0.01	0.08	0.04
28. C_9O_3 (carbo-[3]oxocarbon) (D _{3h} iso1) $\rightarrow$ C_9O_3 (D _{3h} iso2)	28.61	0.00	-0.04	22.88	-0.03	-0.04
29. n-Octane $\rightarrow$ 2,2,3,3-tetramethylbutane	10.61	0.64	0.00	8.12	0.13	0.00
	MAE	0.19	0.06	MAE	0.05	0.05
	ME	0.07	0.02	ME	0.02	0.02
	MRE [%]	0.11	0.06	MRE [%]	0.05	0.06
	RMS	0.27	0.08	RMS	0.07	0.06
	MAX	0.64	0.17	MAX	0.19	0.13

MAE = mean absolute error, ME = mean error, MRE = mean relative error, RMS = root mean square error, MAX = maximum absolute error.

~0.1 kcal/mol, the desired target value. For B3LYP calculations, that only feature 20% exact exchange, the additional error due to the COSX approximation is even negligible and both, RIJDX as well as RIJCOSX, reproduce the reference reaction energies to an accuracy of 0.05 kcal/mol. Thus, for all intents and purposes, the RIJCOSX and RIJDX approximations can be safely used for chemical applications.

4.6. Scaling behavior

The scaling behavior with respect to system size was investigated for a variety of systems shown in Fig. 3. The alkane chains

were constructed in a standard geometry with perfect tetrahedral angles and default bond distances. The DNA basepairs (GC pairs) and glycine chains were constructed using Insight II. The water clusters were initially arranged on a regular grid and then subjected to several 10 ps MD runs with the MM2 force field. The lowest energy arrangement was taken as input for the next MD run and runs were repeated until there was no lower energy arrangement found.

The only system for which we have obtained evidence for linear scaling were the alkane chains were for 9–13 carbons the time for the formation of a single exchange matrix based on the converged density matrix behaves indeed linearly (see Fig. 4).

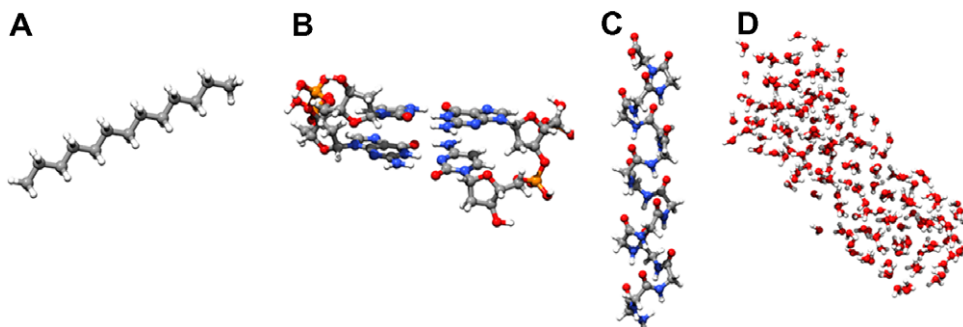


Fig. 3. Systems used for evaluating the scaling behavior of the COSX algorithm. A: linear hydrocarbon chains $\text{H}_3\text{C}-(\text{CH}_2)_n-\text{CH}_3$ B: DNA base pairs (monomer vs. dimer), C: glycine alpha-helices $(\text{gly})_n$ D: water clusters. Except for (A), all calculations are done with the SV basis set.

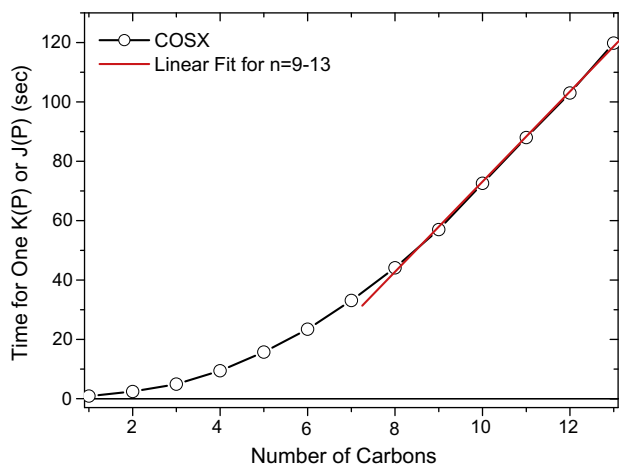


Fig. 4. Scaling behavior of the COSX approximation for linear carbon chains $\text{H}_3\text{C}-(\text{CH}_2)_{n-2}-\text{CH}_3$. Shown is the time for one exchange matrix formation with the converged density. All calculations were done with the SV(P) basis set. Approximate linear scaling is observed for $n \geq 9$.

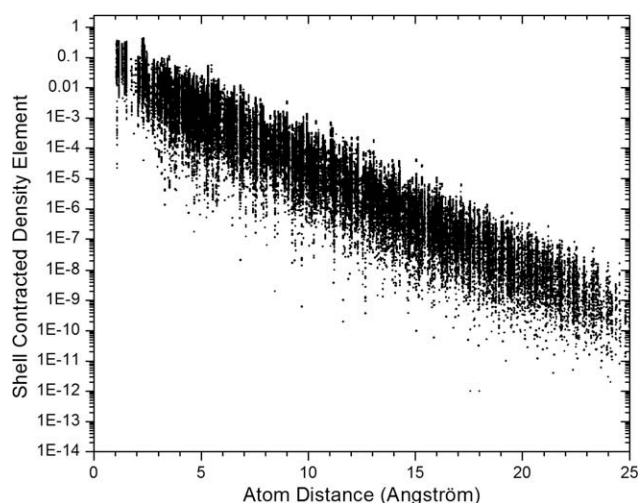


Fig. 5. Logarithmic plot of the shell contracted density matrix $P(K,L)=\max(|P(k,l)|, |P(l,k)|)$, $k \in K, l \in L$ versus the distance of the centers of the two orbital shells. The density is a converged RHF density with the SV basis set.

In all other systems, we have observed essentially exact quadratic scaling. Since the COSX algorithm, by construction, should be linearly scaling, we were curious how this absence of linear scaling arises. A close inspection reveals that this is due to the fact that the secondary shell list did not become constant for a given batch of grid points in the investigated systems. A point in case is shown in Fig. 5 where the shell contracted density matrix is plotted as a function of interatomic distances for $(\text{gly})_{15}$. Consistent with Kohn's conjecture, the values drop roughly exponentially with distance – but with a fairly small decay constant. Hence, a density matrix prescreening value as small as 10^{-10} (that is used in the present shell list construction procedure) is only reached after about 25 Å. Thus, even in this fairly large molecule the P-junctions are such that almost no shell is eliminated from the secondary shell list. Thus, it takes much larger systems to reach the linear scaling regime than the ones investigated here. This changes in later SCF iterations since the difference density matrices quickly become much smaller. Therefore, the exchange construction in the full SCF procedure shows better than quadratic scaling.

Furthermore, the detailed inspection revealed that a second factor contributes to the late onset of linear scaling. This factor is a design flaw of our present numerical integration procedure. It is designed such that the points included in a given batch of grid points are too far apart such that the shell lists that contribute to a given batch are undesirably large. Thus, there is considerable room for improvement of the present COSX algorithm that we are presently investigating. We are confident that the linear scaling regime can be reached much more quickly than what was realized in this work without further loss of accuracy.

4.7. Benchmark results: accuracy of geometries

Finally, we have tested the accuracy of the geometries predicted with the RIJDX and RIJCOSX approximations in RHF calculations with different basis sets on some representative medium sized molecules (Fig. 6).

The results collected in Table 4 demonstrate that for the RIJDX approximation the mean absolute deviations in bond distances and bond angles are consistently less than 0.01 pm and 0.01° relative to the parent RHF approximation. Thus, for all intents and purposes the Split-RI-J method does not introduce any deviations from the canonical results. The errors of the RIJCOSX approximation are about a factor of 10 larger but are still small and in the same range as observed by Friesner and co-worker for the pseudo-spectral method [67]. Bond distances deviate by less than 0.1 pm from their parent values. Bond angles typically deviate by 0.2° from their par-

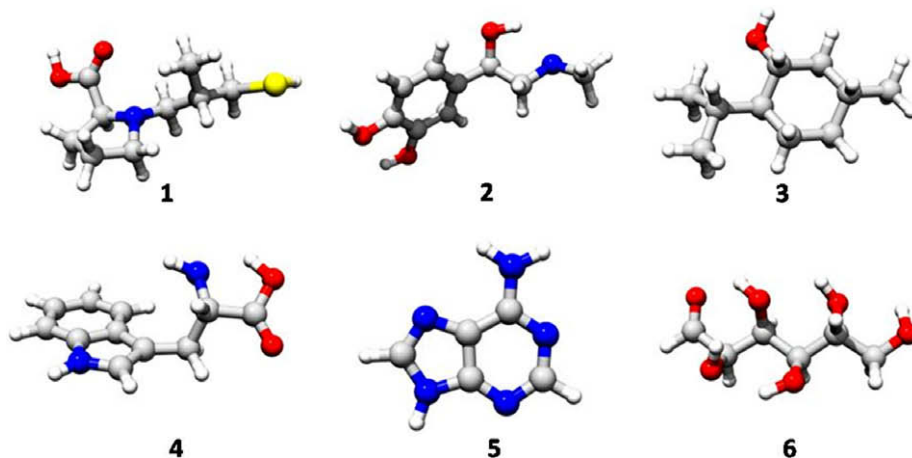


Fig. 6. Systems used for testing the accuracy of geometry optimizations (1 = Captopril, 2 = Epinephrine, 3 = Menthol, 4 = Tryptophane, 5 = Adenine, 6 = D-Glucose).

Table 4

Statistical analysis of the errors in the optimized geometries for molecules 1–6 from the RIJDX and RIJCOSX approximations. Errors are given relative to the parent direct SCF method. All calculations were done with the TZVP basis set (distances in pm, angles are in degrees).^a

		$ \Delta_{\text{mean}}^{\text{distance}} $	$ \Delta_{\text{min}} - \Delta_{\text{max}} ^{\text{distance}}$	$ \Delta_{\text{mean}}^{\text{angle}} $	$ \Delta_{\text{min}} - \Delta_{\text{max}} ^{\text{angles}}$
1.	RIJDX	0.008	0.05	0.012	0.10
	RIJCOSX	0.084	0.42	0.169	0.97
2.	RIJDX	0.004	0.02	0.006	0.04
	RIJCOSX	0.074	0.27	0.130	0.72
3.	RIJDX	0.005	0.02	0.002	0.02
	RIJCOSX	0.076	0.55	0.177	1.57
4.	RIJDX	0.007	0.03	0.005	0.03
	RIJCOSX	0.083	0.55	0.236	1.26
5.	RIJDX	0.004	0.02	0.004	0.02
	RIJCOSX	0.043	0.15	0.065	0.33
6.	RIJDX	0.002	0.02	0.004	0.02
	RIJCOSX	0.072	0.50	0.084	0.55

^a $N(\text{distances})$ for molecules 1–6 = 30,26,31,38,16,23 $N(\text{angles})$ for molecules 1–6 = 56,42,61,46,24,38. Errors in final total energies are for RIJDX = −0.5, −0.4, −0.3, −0.4, −0.3, −0.4 kcal/mol and for RIJCOSX = −0.1, 0.0, 0.5, 0.1, −0.1, 0.2 kcal/mol.

ent value. Only in the case of very floppy bond or dihedral angles (e.g. those of H_2O_2 and N_2H_4) deviations of about up to 1° have been found. However, the deviations of the final total energies from the canonical values remain well below 1 kcal/mol. We argue that these very limited errors are certainly tolerable relative to the other sources of error that are necessarily introduced into any quantum chemical calculation on larger molecules.

4.8. Benchmark results: parallelization

We have implemented a straightforward parallel version of the RIJCOSX algorithm into the ORCA program. The parallelization is realized over batches of grid points. In order to achieve reasonable load balancing, batches of grid points are distributed over processors such that all processors work on all parts of the molecular system to nearly the same extent. Since in ORCA batches are associated with atoms in the input order, this is most easily realized if each processor works only on each $N_{\text{CPU}}^{\text{batch}}$ where N_{CPU} is the number of processes. Only a single gather operation is then performed at the end of the COSX integration loop. Hence, the communication overhead is negligible.

In Fig. 7 the efficiency of the implementation is demonstrated for a relatively large example (Vancomycin/TZV(2d,p), 176 atoms,

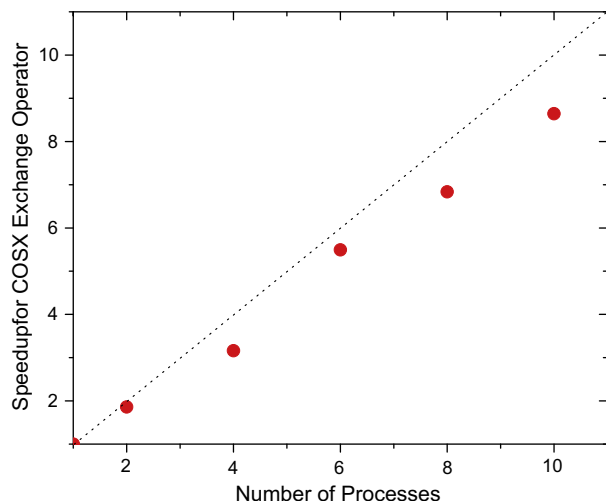


Fig. 7. Parallelization efficiency of the COSX algorithm with Vancomycin/TZV(2d,p) as example. Plotted is the time for one COSX exchange matrix build with the converged density matrix relative to the time taken for the same operation by a single process.

2880 basis functions). As is apparent from Fig. 7, the scaling behavior of the parallel RIJCOSX code is rather good up to six processors (speedup of 5.6) and still acceptable for 10 processors (speedup of 8.6). For larger numbers of processors the system is still too small and the speedup markedly deteriorates to ~ 11 for 16 processes. The results depend somewhat on the computational architecture and the system load.

5. Summary and discussion

In the present work, we have described an efficient approximation to the nonlocal Hartree–Fock exchange. We have demonstrated that the algorithm is: (a) efficient (speedups of up to 60 relative to the traditional implementations were observed with the largest savings realized for basis sets dominated by high angular momentum functions); (b) accurate (average errors <0.3 kcal/mol in total and relative energies as well as negligible errors in geometries); (c) robust and general (independent of the molecular system and the basis set); (d) potentially linear scaling (but actual linear scaling has hardly been reached in the present implementation); (e) efficiently parallelized. The RIJCOSX approximation has been used for almost one year in our research group in many application studies on large open-shell transition metal complexes, their magnetic properties and their reaction mechanisms. No disappointing result compared to the approximation free calculations has thus far been found. We are therefore confident that our default grids and cut-offs are robust. The implementation will be made publically available in the ORCA package in the near future. Despite this success, it is also clear that the treatment of the exchange term is usually still the dominant part of the Fock matrix construction if the Split-RI-J approximation is used for the Coulomb term and research into even more efficient approximations must continue.

Conceptually the COSX algorithm described here is based on semi-numerical integration. The numerical integration aspect allows maximum use of locality to be made while the analytic integration accurately deals with the Coulomb singularity. One may well think of the COSX algorithm as a simplified version of Friesner's pioneering pseudo-spectral method [36–38,67–69,73]. We accept to introduce slightly larger integration grids with the concomitant penalty in computational efficiency and receive in return the generality and simplicity that we have aimed for. The combination of the RI approximation with semi-numerical integration techniques appears to be an obvious alternative for quantum chemical algorithm design but does not appear to have been pursued previously. This is particularly obvious for second-order perturbation calculations such as RI-MP2, RI-SCS-MP2 or double hybrid DFT calculations. All of these methods require large basis set in order to achieve accuracy. Hence the SCF step is usually highly dominant despite its formally lower scaling and therefore all these methods profit enormously from the RIJCOSX algorithm. Comparison of COSX to the existing RI methods for approximately computing the exchange term or the linear scaling exact exchange procedures reveals that the RIJCOSX method is at least highly competitive. For small molecules with highly saturated basis sets the RI-K algorithm by Weigend [63] is likely to be competitive or even superior to RIJCOSX because the efficiency of any pre-screening algorithm will be very limited and in this case the BLAS level 3 procedures on which the RI-K algorithm is based will show excellent performance. However, since RI-K scales as $O(N^4)$ it will be slower than RIJCOSX for larger molecules no matter whether large or small basis sets are used.

Another place where the RIJCOSX approximation could prove useful is the calculation of the exact exchange functional as in the optimized effective potential or localized Hartree–Fock procedures. However, SCF calculations are only one of the places where

exchange type contractions occur in quantum chemistry. Friesner and co-workers have already explored a significant part of the possible remaining applications of semi-numerical techniques [70,71,76,77]. Additional implementations are currently underway in our laboratory and will be reported in future studies. The combination of the RI and COSX techniques, as pursued here, appears to be a very attractive choice under many circumstances.

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Appendix A. Supplementary material

Geometries of the molecules in Tables 1–3. Supplementary data associated with this article can be found, in the online version, at doi:10.1016/j.chemphys.2008.10.036.

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